CMP6202 Artificial Intelligence and Machine Learning

Simple Coursework Pipeline Example (Classification)

```
#to manipulate data as a dataframe
import pandas as pd
#to preprosess data
from sklearn import preprocessing
#to visualise data and results
import matplotlib.pyplot as plt
#to split data into training and test datasets
from sklearn.cross_validation import train_test_split
#to model gaussian naive bayes classifier
from sklearn.naive_bayes import GaussianNB
#to calculate the accuracy score of the model
from sklearn.metrics import accuracy_score


#import dataset
data = pd.read_csv('FileName.csv')

#data understanding, visualisation & pre-processing: see Session 3
#use LabelEncoder() from scikit-learn library to convert string feature values to numbers
#use value_counts() from pandas library to find the distribution of unique values for each (or some) features
#use matplotlib.pyplot.hist() to visualise this distribution

#data slicing
features = data.iloc[:,:8]
target = data.iloc[:,8]
features_train, features_test, target_train, target_test = train_test_split(features,
                                            target, test_size = 0.33, random_state = 10)
#building model using Guasian Naive Bayes algorithm
model = GaussianNB()
model.fit(features_train, target_train)
target_pred = model.predict(features_test)

#print accuracy score
print(accuracy_score(target_test, target_pred, normalize = True)*100)

####Extras####
#cross-validation – see lab 4
#confusion matrix (classification only) – see lab 4
#use another metric (e.g. mean_squared_error for regression) to estimate model performance – see lab 4
#use GridSearchCV for tuning model hyperparameters
```